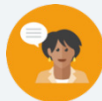


1.10 Solving Inequalities – Part 2

Lesson Introduction

 Benchmark	MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.		 Learning Target	I can solve inequalities and represent the solutions on a number line.
 Benchmark Coherence	Building On: MA.7.AR.2.1 Write and solve one-step inequalities in one variable within a mathematical context and represent solutions algebraically or graphically.		Working Toward: MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.	
 Essential Questions	- How can you solve a two-step inequality that includes distributive property? - How do you graph the answer to an inequality? - What are the different ways you can solve an inequality?			
 Lesson Narrative	In this lesson, students continue to build understanding of solving and graphing inequalities. Students work in pairs to match inequalities, solutions, and graphs in a card sort. Students deepen their understanding by solving given inequalities and creating representations of the solutions using number lines. This understanding is extended to word problems, as students apply their knowledge to create inequalities to represent a real-world context and then use the inequalities to find the solution and describe its meaning in the context of the problem.			
 Component Summary	1.10.1 Warm-Up (~5 min.)	Create an inequality to produce given answer (SE p. 56)	1.10.3 Your Turn! (10-15 min.)	Solve and graph inequalities (SE p. 57)
	1.10.2 Try It (10-15 min.)	Match inequalities to solutions (SE p. 56)	1.10.4 Collaboration (10-15 min.)	Solve and discuss real-world word problems (SE p. 57)
	1.10.5 Wrap-Up (~5 min.)	Students summarize their learning (SE p. 59)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Inequality - Discount		- Cards for 1.10.2 Try It (Optional) Copies of each activity (Optional) Area for centers (Optional) Highlighter (Optional) Paper	
		Additional Preparation		N/A

 Teacher Tips	Common Misconceptions If a problem contains distributive property, students may only distribute to the first term in the parentheses and not the rest of the terms.	Things to Consider - These are all independent activities. Consider creating a quick method for students to ask questions. - Consider using each activity as a center and have students rotate centers. - The real-world problems are an extension of the benchmark. Consider students who may struggle with such an extension and instead utilize a small group as a part of your rotations where these students can receive help.
	Literacy and Language Routines Three Reads 1.10.4 Collaboration presents students with two real-world scenarios in which they must create and solve inequalities before interpreting the solution in the context of the problem. Engage the class in a Three Reads routine to ensure students have a firm grasp on what is happening in each problem and can articulate the information they know and what they are looking for.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Context 1.10.4 Collaboration provides an opportunity for students to engage with inequalities in the context of a real-world scenario, encouraging students to connect the content they have learned in the last three lessons to their own lives. This is an opportunity to build engagement and deepen understanding of inequalities and their purpose.
	 Differentiate Instruction 1.10.2 Try It is a card match activity. Consider ways in which this activity can be differentiated to support engagement for all students. Pairing students based on proficiency levels can provide additional entry points for learners to engage with the content. Consider pairing students who have demonstrated proficiency with students who are struggling with the content to encourage mathematical discourse that can lead to deeper understanding as one “expert” models and explains strategies to their “student.” Alternatively, pairing students of similar proficiency levels can allow for some pairs to work independently while the teacher circulates to support the pairs of students who are still struggling with the material or pulls a small group to provide more scaffolding.	
Lesson Notes:		

1.10.1 Warm-Up: Creating Inequalities

Component Overview: In this Warm-Up, students will create their own inequality using integers from -8 to 8 . The inequality created will need to produce an answer of $x > \frac{2}{3}$.



1.10.1 Warm-Up: Creating Inequalities

- Using the integers -8 to 8 at most one time each, create an inequality with solutions of $x > \frac{2}{3}$. There is more than one solution.

$$\boxed{-6} x < \boxed{-4}$$

Answers will vary.



If students finish early, have them create another inequality that results in the given solution.

1.10.2 Try It: Card Match

Component Overview: In this Try It, students will match the inequality and solution to the graphed answer.



1.10.2 Try It: Card Match

1. Work with your partner to match the provided inequality and solution cards to each graph in the table.

Graph	Inequality	Solution
	$-\frac{2}{3}(x + 6) \geq 3$	$x \leq -10.5$
	$\frac{1}{4}(6x - 8) \geq -0.5$	$x \geq 1$
	$1.75x + 8.2 > 2.95$	$x > -3$
	$\frac{12.7 + 2.6x}{-2} \leq 7.3$	$x \geq -10.5$
	$-15 \geq -5x - 20$	$x \geq -1$
	$-3.5 < -x - 6\frac{1}{2}$	$x < -3$



If utilizing centers, consider having students with the strongest understanding begin with 1.10.4 Collaboration. Students who are struggling should start on 1.10.3 Your Turn!. If working with students who are struggling, consider using 1.10.3 Your Turn!. Utilize the intended flexibility of this lesson to best suit the needs of your students.



This card sort provides a variety of entry points to engage all learners. Engaging students in a card sort provides a kinesthetic entry point for students as they physically interact with the content. The graphs provide a visual entry point for students as they work through and solve each inequality, while partner work provides auditory entry points through discussion and discourse.

1.10.3 Your Turn!

Component Overview: In this Your Turn!, students will solve two-step inequalities and graph the solution.



1.10.3 Your Turn!

1. Solve each inequality and represent the solutions on the number line.

a. $4x - 5 \geq -25$ $x \geq -5$	b. $-1.2 < \frac{y-4}{5}$ $y > -2$
c. $0.2n + 1 \geq -1\frac{3}{5}$ $n \geq -13$	d. $-\frac{1}{2}(-3a + 8) \leq -15$ $a \leq -7\frac{1}{3}$
e. $0.4(6 - 5.2x) > -8$ $x < 5$	f. $\frac{2}{3}a - 5 > 1$ $a > 9$
g. $-3 \leq 4.5 - b$ $b \leq 7.5$	h. $-11 - \frac{y}{4} < -9$ $y > -8$



Consider providing graphs that are already numbered for those who need a visual of where to start when graphing their answer.

1.10.4 Collaboration: Solving Real-World Inequalities

Component Overview: In this Collaboration, students will work with a partner to solve and discuss real-world inequality problems. Students will need to determine what the numbers and parts of the inequality represent. Encourage ongoing discussion during this activity to help students synthesize information.



1.10.4 Collaboration: Solving Real-World Inequalities

Work with your partner to determine the meaning of each inequality in terms of the given situation.

1. Sasha wants to save at least \$300 to purchase a new gaming system. She asked Victor to help her figure out how much she would need to save each month if she has already saved \$50 and would like to buy a system in 6 months. Victor wrote the inequality $6m + 50 \geq 300$, where m represents the dollar amount Sasha must save each month.
 - a. Discuss what the values 6 and 50 represent in the inequality.
6 represents the number of months she has to save the money. \$50 represents the amount she has already saved.
 - b. In terms of the situation, what are the limitations on the possible values of m ?
 m has to be a positive number.
 - c. Solve the inequality.
 $m \geq 41.67$
 - d. Explain what the solution means in terms of the situation.
Sasha needs to make \$41.67 or more to get enough money in 6 months.



Three Reads

Engage students in a Three Reads routine as they complete questions 1 and 2 of this component. This routine helps ensure students understand what information is being presented as well as what the questions are asking them to do. Consider reading the questions aloud to the class first, then having students read through them a second and third time. Students should be able to articulate what Sasha wants in each question.



Consider providing sentence stems to help frame thinking and to fully answer each part of the question.

"_____ represents the number of months she has to save the money. _____ represents the amount she has already saved."

" M has to be a _____ number."

"The solution means Sasha needs to make _____ to have enough money in six months."



Students may not understand that Sasha needs to make at least \$41.67, meaning she can make that amount or any amount more than that. Guide student thinking through questioning.

What does the $\geq$ mean in this context?

Is there just one amount Sasha needs to make the inequality true?

If Sasha makes \$41.67, will she have enough money in six months? Why or why not?

1.10.4 Collaboration: Solving Real-World Inequalities (continued)

2. Sasha received \$100 for her birthday and hopes to buy 4 video games for her gaming system. Each game usually costs \$40, but she's hoping she can find the games on sale this week. She writes the inequality $4(40 - d) \leq 100$, where d represents the dollar amount of the discount on the video games.
 - a. What does the expression $40 - d$ represent?
It represents the \$40 the games cost minus the discount.
 - b. Solve the inequality.
 $d \geq 15$
 - c. Explain what the solution means in terms of the situation.
The discount, d , needs to be at least \$15.00 for Sasha to afford the 4 games.
 - d. Do you think it is likely that Sasha will be able to purchase the 4 video games she wants?
Answers will vary. I think if she finds a coupon, she may be able to purchase them. I think she may have a hard time finding that many games that heavily discounted.



Encourage students to engage in discussion with their partners before completing questions 2c and 2d. Some students may struggle to explain the solution in the context of the situation. Verbalizing before writing can help students process what is happening in the problem.

1.10.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



1.10.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about how the process for solving an inequality is **different** than solving an equation.

Answers will vary.



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities. For example, Desmos offers “Inequalities on a Number Line,” which can be excellent extra practice for students who may have struggled through this lesson or want to continue practicing their skills.